



Nebraska Public Power District

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10 CFR 50.73

NLS2021034
May 17, 2021

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2021-002-00
Cooper Nuclear Station, Docket No. 50-298, DPR-46

Dear Sir or Madam:

The purpose of this correspondence is to forward Licensee Event Report 2021-002-00.

There are no regulatory commitments contained in this letter.

Sincerely,

John Dent, Jr.
Vice President and
Chief Nuclear Officer

/jo

Attachment: Licensee Event Report 2021-002-00

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via IRIS entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Cooper Nuclear Station	2. Docket Number 05000 298	3. Page 1 OF 4
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4. Title Valve Test Failures Result in Condition Prohibited by Technical Specifications
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5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
03	20	2021	2021	- 002 -	00	05	17	2021	Facility Name	05000

9. Operating Mode 1	10. Power Level 100
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)				
10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	
☐ Other (Specify here, in Abstract, or in NRC 366A).				

12. Licensee Contact for this LER	
Licensee Contact Linda Dewhirst, Regulatory Affairs and Compliance Manager	Phone Number (Include Area Code) (402) 825-5416

13. Complete One Line for each Component Failure Described in this Report									
Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
B	SB	RV	T020	Y					

14. Supplemental Report Expected					15. Expected Submission Date		
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				Month	Day	Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On March 18 and March 20, 2021, two Target Rock safety relief valve (SRV) pilot assemblies, removed during Refueling Outage 31, failed to lift within Technical Specifications (TS) lift setpoint requirements. The pressure setpoint of the first failed pilot assembly was 1080 psig; the SRV pilot assembly lifted at 1191.9 psig. The pressure setpoint of the second failed pilot assembly was 1100 psig; the SRV pilot assembly lifted at 1180.7 psig. Three subsequent lifts were performed for the failed SRV pilot assemblies and the results were within the TS pressure setpoint tolerance.

The cause of the failure was determined to be corrosion bonding.

Although the setpoint lift pressures of the SRV pilot valve assemblies exceeded TS requirements, an analysis of this event concluded the thermal limits, and American Society of Mechanical Engineers overpressure transients, did not exceed allowable limits. Public safety was not at risk. Safety of plant personnel, and plant equipment, was not at risk.

A TS Amendment to TS 3.4.3 will be submitted that will address the failure to meet the TS criteria for the Target Rock two-stage SRVs. Following submittal, a tracking mechanism will be established for TS change approval and implementation.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Cooper Nuclear Station	05000- 298	YEAR	SEQUENTIAL NUMBER	REV NO.
		2021	- 002	- 00

NARRATIVE**PLANT STATUS**

Cooper Nuclear Station (CNS) was in Mode 1, Power Operation, at 100 percent power, when the event was discovered; i.e., March 20, 2021.

BACKGROUND

The pressure relief system includes three American Society of Mechanical Engineers (ASME) code safety valves (SV) [EIS: SB] and eight safety relief valves (SRV) [EIS: RV], all of which are located on the main steam lines [EIS: SB] within the drywell [EIS: NH], between the reactor vessel [EIS: RPV] and the first main steam isolation valve [EIS: ISV]. The SVs provide protection against over pressurization of the nuclear system and discharge directly into the interior space of the drywell. The SRVs discharge to the suppression pool and provide three main functions: overpressure relief operation to limit the pressure rise and prevent safety valve opening, overpressure safety operation to prevent nuclear system over pressurization, and depressurization operation (opened automatically or manually) as part of the emergency core cooling system [EIS: BJ, BM, BO].

Technical Specification (TS) Limiting Condition for Operation 3.4.3 requires the safety function of seven SRVs and three SVs to be operable. The nominal set pressure tolerances for these valves are established in CNS TS Surveillance Requirement (SR) 3.4.3.1.

The SRVs installed at CNS are Target Rock Model 7567F, two-stage, pilot-actuated valves with pilot assemblies comprised of Stellite 21 pilot discs and Stellite 6B pilot body seats. The pilot assemblies had been in continuous service since installation in Refueling Outage (RE) 30, (Fall of 2018).

Corrosion bonding is an industry known problem with Target Rock Model 7567F two-stage relief valves in which the pilot valve bonds/sticks to the in-body seat, resulting in an increased pressure needed to lift the valve. After the corrosion bonding is "broken" following the initial lift, the subsequent lifts are within tolerance.

EVENT DESCRIPTION

In the Fall of 2020, four main SRV body assemblies and four SRV pilot assemblies were removed during RE31. In March 2021, the SRVs were as-found tested at National Technical Systems Laboratories.

The pressure setpoint for SRV pilot assembly serial number 378 is 1080 psig. The TS SR 3.4.3.1 as-found limit acceptance is 1080 +/- 3%. The first actual lift pressure of this SRV pilot assembly was recorded at 1191.9 psig, 10.36% above the pressure setpoint. A second, third, and fourth lift was performed and the results were 1103 psig, 1084 psig, and 1082 psig, all three within 3% of the pressure setpoint.

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CONTINUATION SHEET**

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The pressure setpoint for SRV pilot assembly serial number 376 is 1100 psig. The TS SR 3.4.3.1 as-found limit acceptance is 1100 +/- 3%. The first actual lift pressure of this SRV pilot assembly was recorded at 1180.7 psig, 7.33% above the pressure setpoint. A second, third, and fourth lift was performed and the results were 1096 psig, 1093 psig, and 1089 psig, all three within 3% of the pressure setpoint.

BASIS FOR REPORT

CNS is reporting this event as an operation or condition prohibited by plant TS per 10 CFR 50.73(a)(2)(i)(B), and also as a condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident as defined under 10 CFR 50.73(a)(2)(v)(D).

SAFETY SIGNIFICANCE

CNS performed an Engineering Report (ER) documenting the impact of the RE31 SRV setpoint failures. The ER determined the ASME overpressure protection and core thermal limits would not have been exceeded. Public safety was not at risk. Safety to plant personnel and plant equipment was not at risk.

Based on the results of the ER, this event is not considered a Safety System Functional Failure nor a loss of safety function and will not be counted as a Safety System Functional Failure to the Nuclear Regulatory Commission (NRC) performance indicator since no loss of safety function occurred.

CAUSE

Corrosion bonding between the pilot disc and the in-body seat.

CORRECTIVE ACTIONS

The following corrective actions have been entered into CNS' corrective action program:

1. Track to completion the Boiling Water Reactor Owners Group industry initiative to develop a Technical Specifications Task Force Traveler for TS 3.4.3, including NRC approval, that would address the failure to meet TS criteria for the Target Rock two-stage SRVs.
2. Submit a TS Amendment to TS 3.4.3 that would address the failure to meet TS criteria for the Target Rock two-stage SRVs. Following submittal, initiate a new CA for TS Change approval that also ensures a follow on corrective action to implement the TS change.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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PREVIOUS EVENTS

Licensee Event Report (LER) 2019-001-00 and LER 2019-001-01 – On March 4, March 5, and March 6, 2019, three of eight Target Rock SRV pilot assemblies failed to lift within TS lift setpoint requirements. The cause was corrosion bonding.

LER 2017-002-00 – In February and March, 2017, one Target Rock SRV pilot assembly failed to lift within TS lift setpoint requirements and another SRV pilot assembly was conservatively considered a failure due to lack of as-found lift pressure test data since it was inadvertently disassembled prior to performing the as-found lift pressure test. There were two causes for the failures; one of the SRV pilot assemblies failed due to corrosion bonding; the other SRV pilot assembly failed due to a lack of a barrier to prevent inadvertent disassembly of the SRV pilot prior to testing.

LER 2015-001-00 and LER 2015-001-01 – On January 26 and February 11, 2015, five of eight Target Rock SRV pilot valve assemblies failed to lift within TS lift setpoint requirements. The cause was corrosion bonding.

LER 2011-005-00 – On June 22, 2011, one of eight Target Rock SRV pilot valve assemblies failed to lift within TS lift setpoint requirements. The cause was corrosion bonding.

LER 2010-001-00 – On January 12, 2010, two of eight Target Rock SRV pilot valve assemblies failed to lift within TS lift setpoint requirements. The cause was corrosion bonding.

LER 2008-002-00 – On July 7 through July 9, 2008, one of eight Target Rock SRV pilot assemblies failed to lift within TS lift setpoint requirements. The cause was corrosion bonding.

LER 2007-002-00 – On February 28 through March 2, 2007, one of eight Target Rock SRV pilot valve assemblies failed to lift within its TS lift setpoint. The failure was a result of sufficient corrosion bonding between the SRV pilot valve assembly Stellite 21 disc and the pilot valve Stellite 6B body seat to cause the SRV pilot valve to lift outside its TS setpoint tolerance.